

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ad57200fafd70e3ef.jsonl`  
- SHA-256: `bbfa266012acb653163bffd01b400a4c9952dd4585af6a150db8b0f741a2c253`  
- Source modified: `2026-06-16T03:25:36+00:00`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:55.280Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer is a single-stage, anchor-free temporal action localization model that combines a multiscale feature pyramid with local self-attention (a Transformer) and a lightweight decoder that classifies every moment in time and regresses the corresponding action boundaries, eliminating the need for separate action proposals."

****Source:**** https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)

****Supporting quote:**** "The model employs "a multiscale feature representation with local self-attention" and "a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries." It operates as a single-stage, anchor-free detector without requiring action proposals."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:58.324Z)

I'll verify this claim about ActionFormer by checking the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:25:00.818Z)

I'll fetch the primary source to verify the claim against the actual paper content.

4. user (2026-06-16T03:25:03.502Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. user (2026-06-16T03:25:09.626Z)

Web search results for query: "ActionFormer single-stage anchor-free temporal action localization multiscale local self-attention decoder classify every moment regress boundaries"

Links: [{"title": "ActionFormer: Transformer-based Action Localization", "url": "https://www.emergentmind.com/papers/2202.07925"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://www.arxiv.org/pdf/2512.01298"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "[PDF] ActionFormer: Localizing Moments of Actions with Transformers | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/ActionFormer:-Localizing-Moments-of-Actions-with-Zhang-Wu/58a3fedc03ab9f5908c077c115eff4c8d2d87660"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}, {"title": "[2512.01298] TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/abs/2512.01298"}]

I found relevant information about ActionFormer. Here's what the search results reveal:

ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries. It is a single-stage anchor-free model for temporal action localization.

The primary components of ActionFormer include a multiscale feature representation, local self-attention for context modeling, and a lightweight convolutional decoder. The local self-attention enhances long-range temporal dependencies essential for distinguishing action boundaries.

ActionFormer has a simple design, falls into the category of single-stage anchor-free method, yet achieves impressive results across several major TAL (Temporal Action Localization) benchmarks including THUMOS14, ActivityNet-1.3, and the more recent EPIC-Kitchens 100. Without bells and whistles, ActionFormer achieves 71.0% mAP at tIoU = 0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:12.256Z)

The ACM page returned 403, but the arXiv version (same paper, ECCV 2022) is accessible. Let me fetch the official arXiv abstract verbatim to confirm the exact wording.

7. user (2026-06-16T03:25:17.370Z)

ActionFormer Analysis

****Abstract (verbatim, shortened):****

"ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries."

****Answers to your questions:****

? ****Multiscale features + local self-attention:**** Yes, explicitly stated in the abstract.

? ****Lightweight decoder for moment classification and boundary estimation:**** Yes, confirmed in the abstract.

? ****Single-stage and anchor-free:**** Yes. The abstract notes it identifies "actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows."

? ****No action proposals or anchor windows required:**** Confirmed?the model operates "without using action proposals or relying on pre-defined anchor windows."

8. assistant (2026-06-16T03:25:25.442Z)

The arXiv abstract (the canonical version of the same ECCV 2022 paper cited at the ACM DOI) confirms every component of the claim verbatim.

Let me verify each element against the abstract:

1. ****"single-stage, anchor-free"**** ? Abstract: "single-shot, without using action proposals or relying on pre-defined anchor windows." Confirmed.
2. ****"multiscale feature pyramid with local self-attention (a Transformer)"**** ? Abstract: "combines a multiscale feature representation with local self-attention." Confirmed. (Local self-attention = Transformer; "pyramid" is the standard term for the multiscale representation ? accurate paraphrase.)
3. ****"lightweight decoder that classifies every moment in time and regresses the corresponding action boundaries"**** ? Abstract: "uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries." Confirmed nearly verbatim ("regresses" = "estimate" via boundary regression heads, accurate).
4. ****"eliminating the need for separate action proposals"**** ? Abstract: "without using action proposals." Confirmed.

The supporting quote in the claim accurately reflects the abstract. The source is a primary, peer-reviewed venue (ECCV 2022, ACM DOI), and the arXiv version matches. Every assertion is directly supported. The only minor rephrasings ("pyramid" for "representation," "regresses" for "estimate") are standard, accurate terminology in the TAL literature. No contradicting evidence found; multiple credible sources (emergentmind, the paper PDF itself) corroborate.

This claim is well-supported, current, and from a primary high-quality source. Not refuted.

9. user (2026-06-16T03:25:32.181Z)

Structured output provided successfully

10. assistant (2026-06-16T03:25:35.934Z)

Claim verified as ****not refuted**** with high confidence. The ActionFormer description matches the primary source abstract (ECCV 2022 / arXiv:2202.07925) on all four components: single-stage anchor-free, multiscale features + local self-attention, lightweight decoder classifying every moment and regressing boundaries, and no action proposals. The minor rephrasings ("pyramid," "regresses") are accurate standard TAL terminology. Source quality is sufficient (peer-reviewed primary), and multiple independent sources corroborate.